

Hadoop MapReduce Word Count on Multiple Files

1. Objective

To create an HDFS directory, upload multiple text files, perform Word Count using Hadoop MapReduce on all files collectively, and identify the top 3 most frequent words from the generated output.

2. Procedure

Step 1: Create a Directory in HDFS

```
hdfs dfs -mkdir /my_data_directory
```

Step 2: Create Three Text Files Locally

```
echo "This is the content of file1.txt" > file1.txt  
echo "This is the content of file2.txt" > file2.txt  
echo "This is the content of file3.txt" > file3.txt
```

Step 3: Upload the Files to HDFS

```
hdfs dfs -put file1.txt /my_data_directory/  
hdfs dfs -put file2.txt /my_data_directory/  
hdfs dfs -put file3.txt /my_data_directory/
```

Step 4: Verify the Uploaded Files

List Files in the Directory

```
hdfs dfs -ls /my_data_directory
```

View File Contents

```
hdfs dfs -cat /my_data_directory/file1.txt  
hdfs dfs -cat /my_data_directory/file2.txt  
hdfs dfs -cat /my_data_directory/file3.txt
```

Step 5: Execute the Word Count Program

```
hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar  
wordcount /my_data_directory /wordcount_output
```

Step 6: View the Word Count Results

```
hdfs dfs -cat /wordcount_output/part-r-000000
```

Step 7: Find the Top 3 Most Frequent Words

```
hdfs dfs -cat /wordcount_output/part-r-000000 | sort -t$'\t' -k2 -nr | head -3
```

3. Source Code

Create HDFS Directory

```
hdfs dfs -mkdir /my_data_directory
```

Create Input Files

```
echo "This is the content of file1.txt" > file1.txt  
echo "This is the content of file2.txt" > file2.txt  
echo "This is the content of file3.txt" > file3.txt
```

Upload Files to HDFS

```
hdfs dfs -put file1.txt /my_data_directory/  
hdfs dfs -put file2.txt /my_data_directory/  
hdfs dfs -put file3.txt /my_data_directory/
```

Verify Uploaded Files

```
hdfs dfs -ls /my_data_directory
```

```
hdfs dfs -cat /my_data_directory/file1.txt  
hdfs dfs -cat /my_data_directory/file2.txt  
hdfs dfs -cat /my_data_directory/file3.txt
```

Execute Word Count

```
hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar  
wordcount /my_data_directory /wordcount_output
```

View Results

```
hdfs dfs -cat /wordcount_output/part-r-000000
```

Get Top 3 Most Frequent Words

```
hdfs dfs -cat /wordcount_output/part-r-000000 | sort -t$'\t' -k2 -nr | head -3
```

4. Screenshots

```
sky@MS97377:~$ hdfs dfs -mkdir /my_data_directory
sky@MS97377:~$ hdfs dfs -ls /
Found 5 items
drwxr-xr-x - sky supengroup 0 2026-07-26 23:28 /demo
drwxr-xr-x - sky supengroup 0 2026-07-26 23:54 /my_data_directory
drwxr-xr-x - sky supengroup 0 2026-07-26 23:31 /output
drwxr-xr-x - sky supengroup 0 2026-07-25 16:40 /tmp
drwxr-xr-x - sky supengroup 0 2026-07-25 16:37 /user
sky@MS97377:~$ echo "This is the content of file1.txt" > file1.txt
sky@MS97377:~$ echo "This is the content of file2.txt" > file2.txt
sky@MS97377:~$ echo "This is the content of file3.txt" > file3.txt
sky@MS97377:~$ ls
'# Add the Docker repository'  docker-compose.yml  generate_sales.py  mapper_monthly_sales.py  reducer_monthly_sales.py  setup_exercise.cql
exercise6                      file1.txt           hadoop-3.3.6.tar.gz  mapper_product_qty.py    reducer_product_qty.py    snap
cassandra-exercise            file2.txt           hadoop-3.3.6.tar.gz.1  prac.txt                 sales_analysis.png        visualize_sales.py
city_sales_results.txt        file3.txt           mapper_city_sales.py   reducer_city_sales.py    sales_data.csv            zennml_project
sky@MS97377:~$ cat file1.txt
This is the content of file1.txt
sky@MS97377:~$ cat file2.txt
This is the content of file2.txt
sky@MS97377:~$ cat file3.txt
This is the content of file3.txt
sky@MS97377:~$ hdfs dfs -put file1.txt /my_data_directory/
sky@MS97377:~$ hdfs dfs -put file2.txt /my_data_directory/
sky@MS97377:~$ hdfs dfs -put file3.txt /my_data_directory/
sky@MS97377:~$ hdfs dfs -ls /my_data_directory
Found 3 items
-rw-r--r-- 1 sky supengroup 33 2026-07-26 23:57 /my_data_directory/file1.txt
-rw-r--r-- 1 sky supengroup 33 2026-07-26 23:57 /my_data_directory/file2.txt
-rw-r--r-- 1 sky supengroup 33 2026-07-26 23:57 /my_data_directory/file3.txt
sky@MS97377:~$ hdfs dfs -cat /my_data_directory/file1.txt
This is the content of file1.txt
sky@MS97377:~$ hdfs dfs -cat /my_data_directory/file2.txt
This is the content of file2.txt
sky@MS97377:~$ hdfs dfs -cat /my_data_directory/file3.txt
This is the content of file3.txt
sky@MS97377:~$ hdfs dfs -rm -r /wordcount_output
rm: '/wordcount_output': No such file or directory
sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /my_data_directory /wordcount_output
2026-07-26 23:59:24,464 INFO client.DefaultHohARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-26 23:59:25,128 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1785087544616_0002
2026-07-26 23:59:26,122 INFO input.FileInputFormat: Total input files to process : 3
2026-07-26 23:59:26,203 INFO mapreduce.JobSubmitter: number of splits:3
```

```

sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /my_data_directory /wordcount_output
2026-07-26 23:59:24,464 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-26 23:59:25,128 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1785087544616_0002
2026-07-26 23:59:26,122 INFO input.FileInputFormat: Total input files to process : 3
2026-07-26 23:59:26,293 INFO mapreduce.JobSubmitter: number of splits:3
2026-07-26 23:59:26,611 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1785087544616_0002
2026-07-26 23:59:26,611 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-26 23:59:27,025 INFO conf.Configuration: resource-types.xml not found
2026-07-26 23:59:27,026 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-07-26 23:59:27,160 INFO impl.YarnClientImpl: Submitted application application_1785087544616_0002
2026-07-26 23:59:27,335 INFO mapreduce.Job: The url to track the job: http://MS97377.localdomain:8088/proxy/application_1785087544616_0002/
2026-07-26 23:59:27,340 INFO mapreduce.Job: Running job: job_1785087544616_0002
2026-07-26 23:59:38,758 INFO mapreduce.Job: Job job_1785087544616_0002 running in uber mode : false
2026-07-26 23:59:38,762 INFO mapreduce.Job: map 0% reduce 0%
2026-07-26 23:59:55,337 INFO mapreduce.Job: map 67% reduce 0%
2026-07-26 23:59:56,360 INFO mapreduce.Job: map 100% reduce 0%
2026-07-27 00:00:07,497 INFO mapreduce.Job: map 100% reduce 100%
2026-07-27 00:00:09,554 INFO mapreduce.Job: Job job_1785087544616_0002 completed successfully
2026-07-27 00:00:09,738 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=213
    FILE: Number of bytes written=1103519
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=441
    HDFS: Number of bytes written=0
    HDFS: Number of read operations=14
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=3
    Launched reduce tasks=1
    Data-local map tasks=3
    Total time spent by all maps in occupied slots (ms)=40855
    Total time spent by all reduces in occupied slots (ms)=9218
    Total time spent by all map tasks (ms)=40855
    Total time spent by all reduce tasks (ms)=9218
    Total vcore-milliseconds taken by all map tasks=40855
    Total vcore-milliseconds taken by all reduce tasks=9218
    Total megabyte-milliseconds taken by all map tasks=41835520
  Job Counters
    Launched map tasks=3
    Launched reduce tasks=1
    Data-local map tasks=3
    Total time spent by all maps in occupied slots (ms)=40855
    Total time spent by all reduces in occupied slots (ms)=9218
    Total time spent by all map tasks (ms)=40855
    Total time spent by all reduce tasks (ms)=9218
    Total vcore-milliseconds taken by all map tasks=40855
    Total vcore-milliseconds taken by all reduce tasks=9218
    Total megabyte-milliseconds taken by all map tasks=41835520
    Total megabyte-milliseconds taken by all reduce tasks=9439232
  Map-Reduce Framework
    Map input records=3
    Map output records=18
    Map output bytes=171
    Map output materialized bytes=225
    Input split bytes=342
    Combine input records=18
    Combine output records=18
    Reduce input groups=8
    Reduce shuffle bytes=225
    Reduce input records=18
    Reduce output records=8
    Spilled Records=36
    Shuffled Maps =3
    Failed Shuffles=0
    Merged Map outputs=3
    GC time elapsed (ms)=1184
    CPU time spent (ms)=10940
    Physical memory (bytes) snapshot=1249882112
    Virtual memory (bytes) snapshot=10178293760
    Total committed heap usage (bytes)=1037041664
    Peak Map Physical memory (bytes)=335822848
    Peak Map Virtual memory (bytes)=2543591424
    Peak Reduce Physical memory (bytes)=255762432
    Peak Reduce Virtual memory (bytes)=2550046720
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
sky@MS97377:~$ hdfs dfs -cat /wordcount_output/part-r-00000
This      3
content   3
file1.txt 1
file2.txt 1
file3.txt 1
is        3
of        3
the       3
the       3
sky@MS97377:~$ hdfs dfs -cat /wordcount_output/part-r-00000 | sort -t$'\t' -k2 -nr | head -3
the       3
of        3
is        3
sky@MS97377:~$

```

5. Conclusion

The experiment successfully demonstrated Hadoop MapReduce processing on multiple input files stored in HDFS. Three text files were uploaded into an HDFS directory and processed collectively using the Word Count application. The generated output provided the frequency of each word across all files, and Linux commands were used to identify the top three most frequently occurring words. This experiment illustrates Hadoop's capability to process data distributed across multiple files efficiently.